Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



Team Details

a. Team name: EWS - GDG Hanoi
b. Team leader name: Dinh Huu Hoang
c. Problem Statement: Banks only flag a loan as bad once it is 30 days past due, which is too late to act, and they cannot see how one borrower's failure spreads to the related companies, guarantors and partners around it.

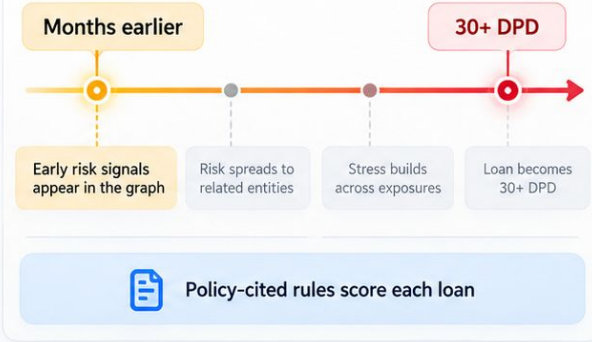
Brief about the idea

See bad loans coming. See risk spreading.

Graph-aware early warning for non-performing loans



Knowledge graph: borrowers · loans · collateral · companies · guarantors · industries



Gemini API · 3 roles

Analyst — explains alerts with verified graph evidence

Portfolio assistant — answers through typed tools

News reader — links English & Vietnamese risk news to the graph

Gemini never sets scores · Names masked before calls

Officer approves → task created → mitigation measured again

Cloud Run: UI + API in one container

Built by Cloud Build · Stored in Artifact Registry

Opportunities

- **How different is it from any of the other existing ideas?**
  - Existing early-warning tools score each loan on its own and use AI as a black box. Early Warning scores loans on a knowledge graph, so it sees risk spreading from related companies, guarantors and partners. Transparent rules set every score, and Gemini only explains, citing graph evidence.
- **How will it be able to solve the problem?**
  - It flags deteriorating loans about 4 months before default, roughly 2 months earlier than the 30-days-past-due rule. Officers can simulate a counterparty default and see the domino effect, and news is turned into graph events. Approved actions become tasks whose effect on risk is measured.
- **USP of the proposed solution**
  - Graph-aware contagion, from detection through to measured action. Officers can make any node default with one click and watch the cascade, and every AI claim is verified against the graph. It is explainable and auditable, with a human in the loop

List of features offered by the solution

What Early Warning offers



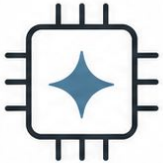
Command Center

Portfolio mix, industry heatmap & warning timeline



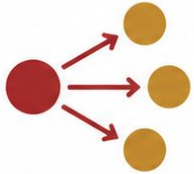
Case Review & Time Travel

Context graph & 6 policy-cited signals, any month



Gemini Analyst

Verified graph evidence; names masked before AI calls



Contagion Lab

Trigger a default and watch the cascade, wave by wave



Contagion Matrix

See which counterparties spread risk by industry



News to Graph

Link English or Vietnamese news to entities; preview impact



Action Board

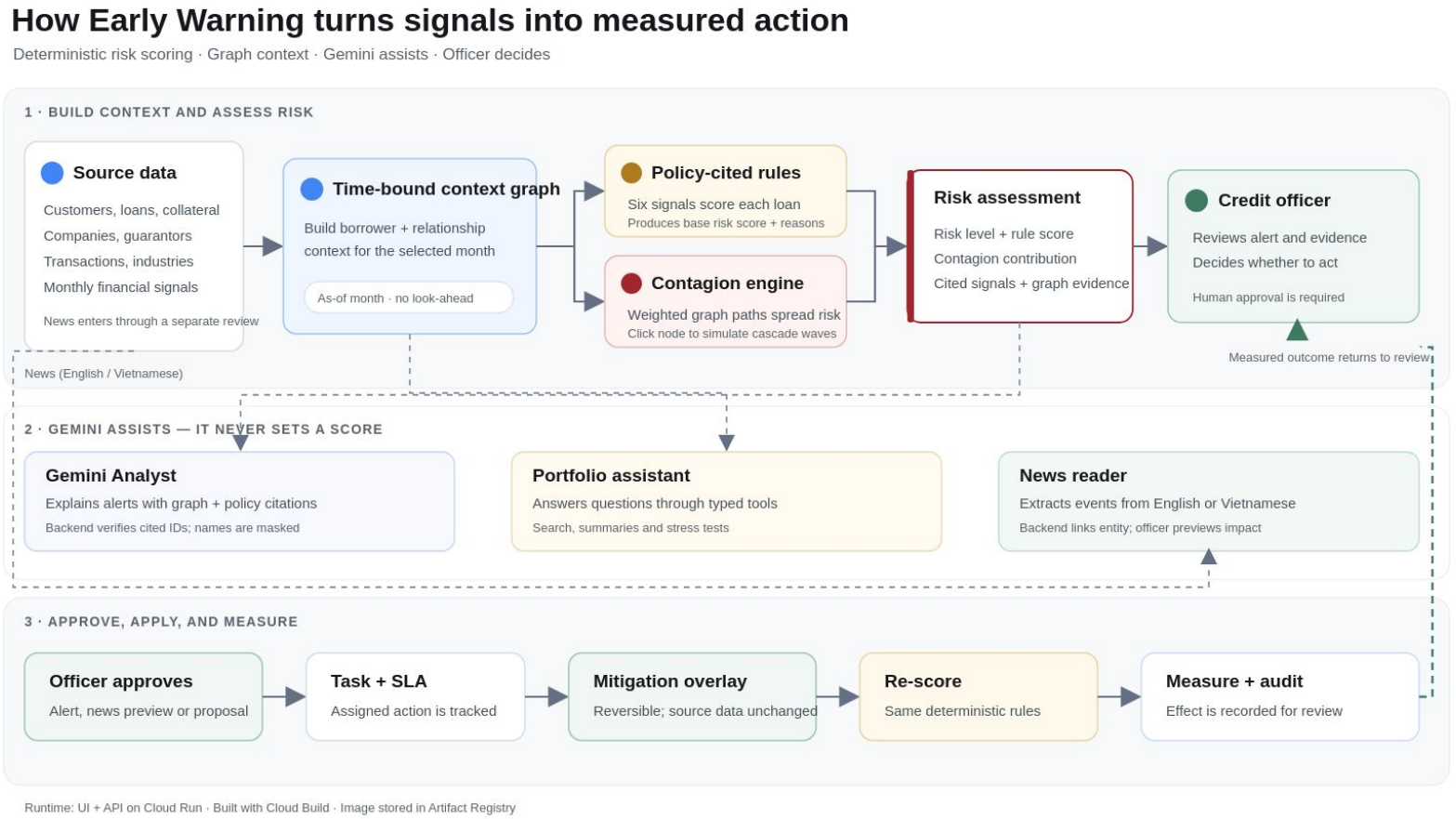
Approved tasks with SLAs; measure mitigation effects



Ask Gemini & Stress Test

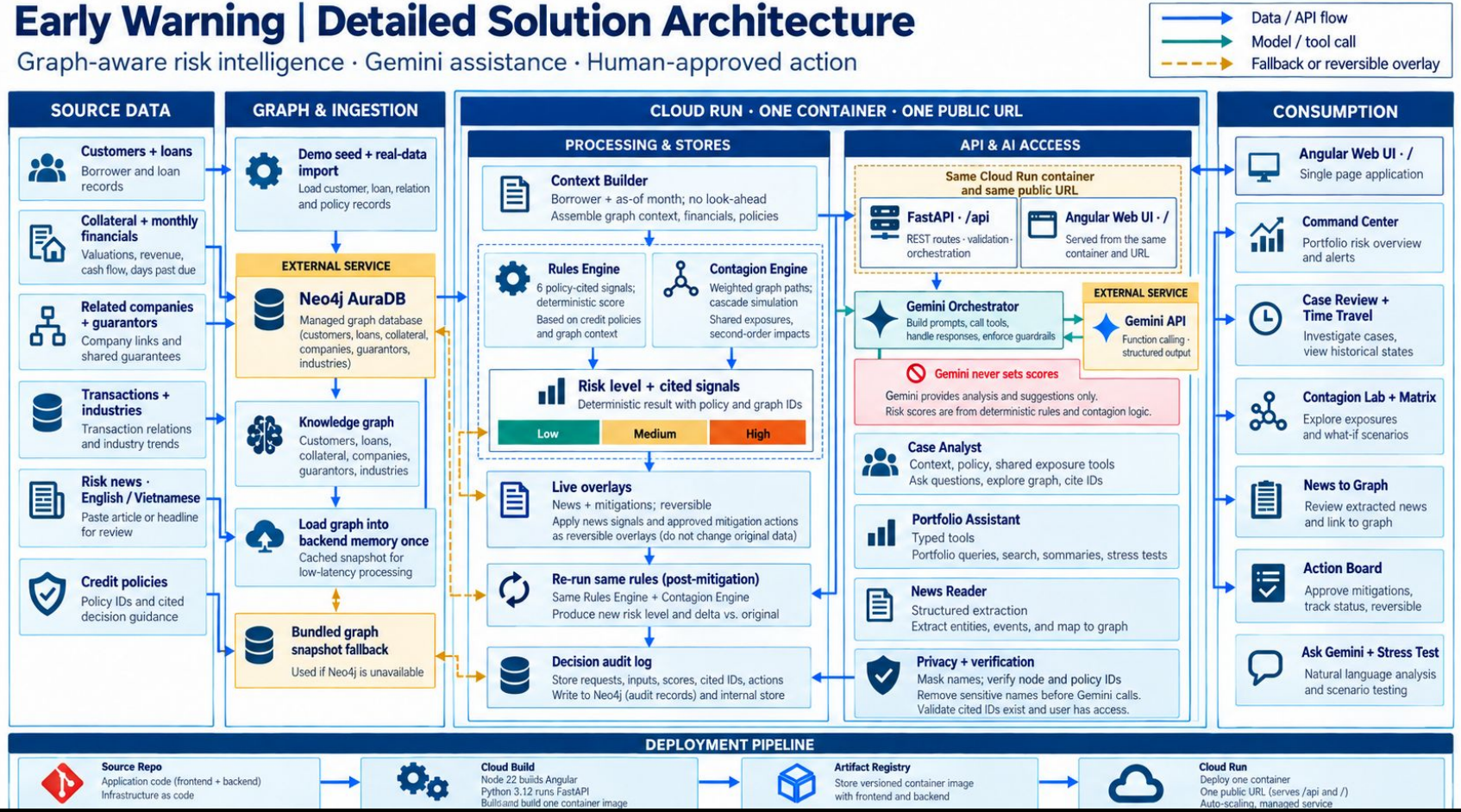
Ask in natural language; shock-test the portfolio

Process flow diagram or Use-case diagram



Early Warning | Detailed Solution Architecture

Graph-aware risk intelligence · Gemini assistance · Human-approved action



Technology Stack | Early Warning

From borrower graph to explainable risk action

1. USER EXPERIENCE

- **Angular 22** — Standalone components · Signals
- **Cytoscape.js** — Interactive knowledge graph
- **TypeScript · RxJS** — Reactive frontend development

KEY OUTCOMES

- Command Center
- Case Review
- Contagion Lab
- Action Board

2. API & APPLICATION

- **Python 3.12** — Runtime for backend services
- **FastAPI + Uvicorn** — REST API at /api
- **Pydantic** — Structured request / response models
- **Angular UI + FastAPI API** — Served from the same container and public URL (Cloud Run)

3. RISK & AI

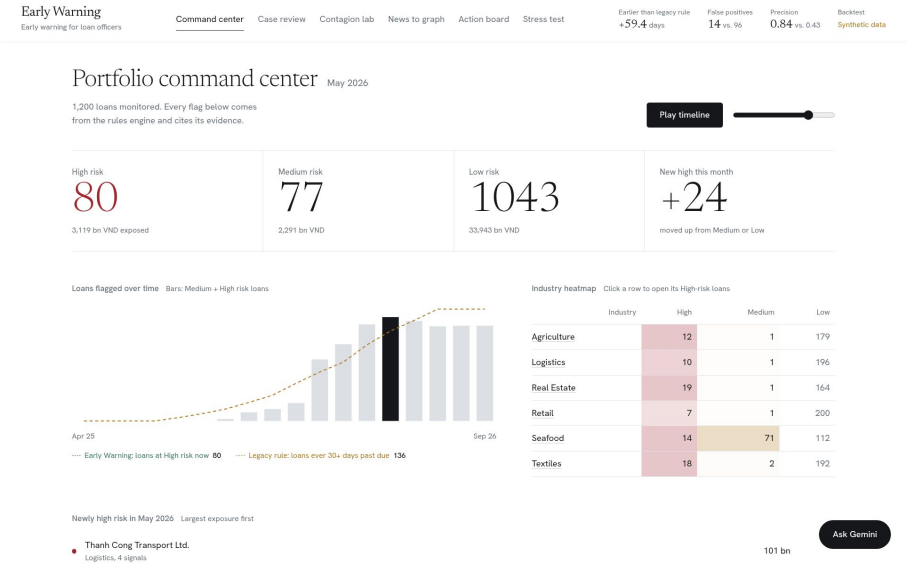
- **Deterministic Python rules** — 6 policy-cited warning signals. Rules own every score
- **Graph contagion engine** — Weighted paths · cascade simulation. Graph-based risk propagation
- **Gemini API · google-genai** — Function calling · structured output. Explains, extracts and suggests
- **Guardrails** — Gemini explains and extracts; it **never scores.**

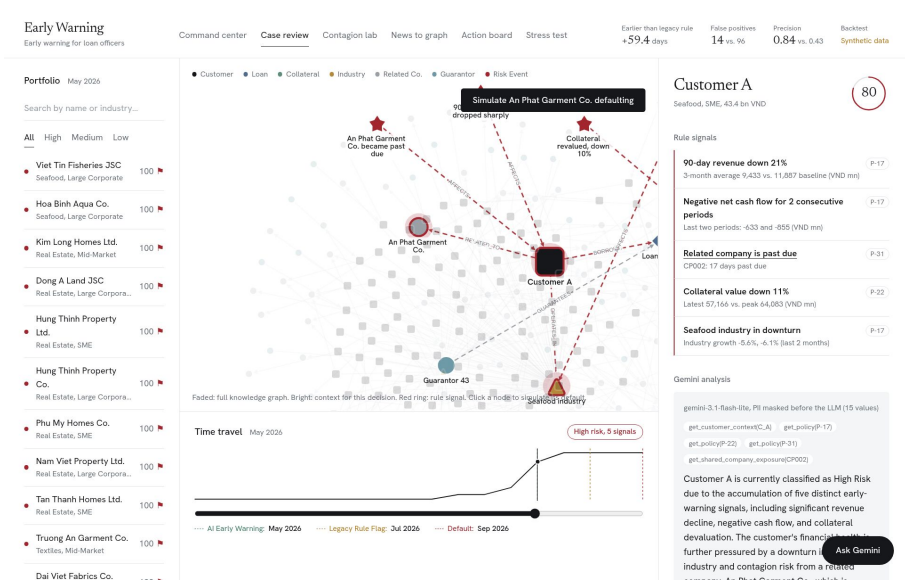
4. DATA & STATE

- **Neo4j AuraDB · Cypher** — Borrower knowledge graph · policies · audit records. Primary graph store (external)
- **In-memory graph + bundled snapshot** — Fast reads · fallback if Neo4j is unavailable
- **Reversible live overlays** — News events · approved mitigations. Apply and revert (non-destructive)

DEPLOYMENT PIPELINE

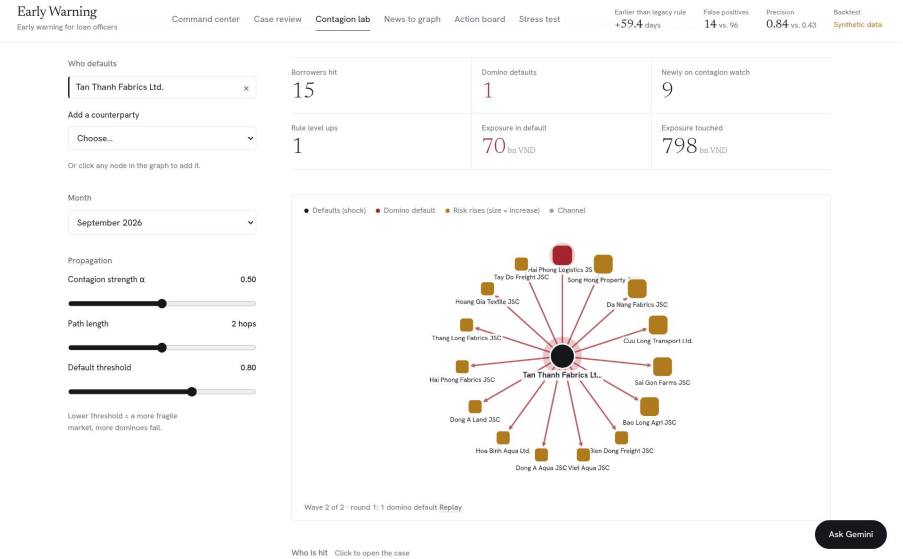
1. **Docker multi-stage build** — Build Angular (Node 22) + FastAPI (Python 3.12)
2. **Cloud Build** — Node 22 / npm · Python 3.12 Build container image
3. **Artifact Registry** — Store versioned container image
4. **Cloud Run** — One container · Angular + FastAPI · one public URL





Command Center

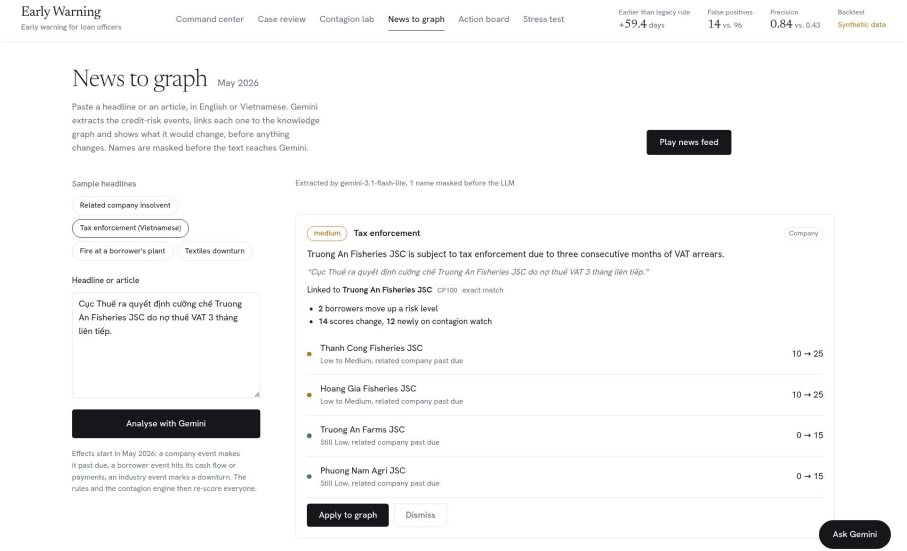
Case review Gemini



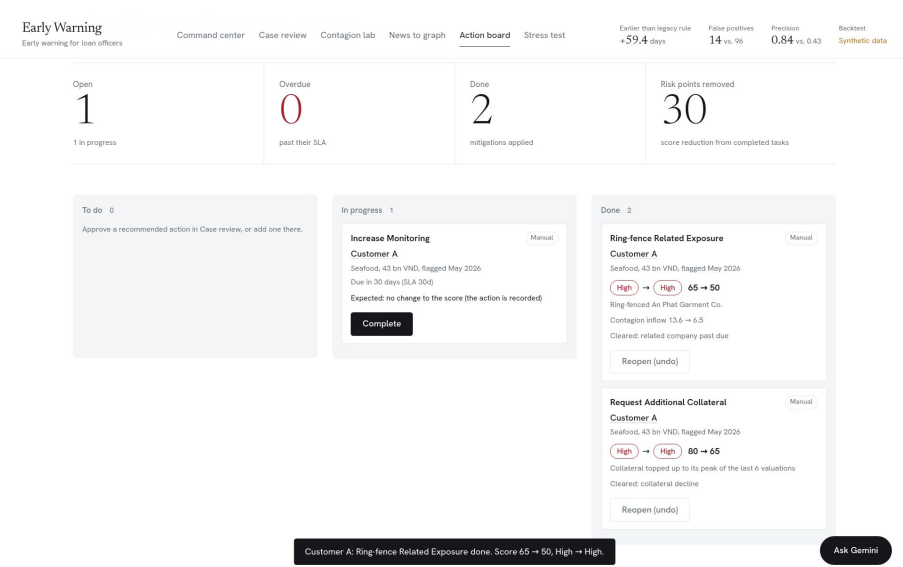
Contagion Lab



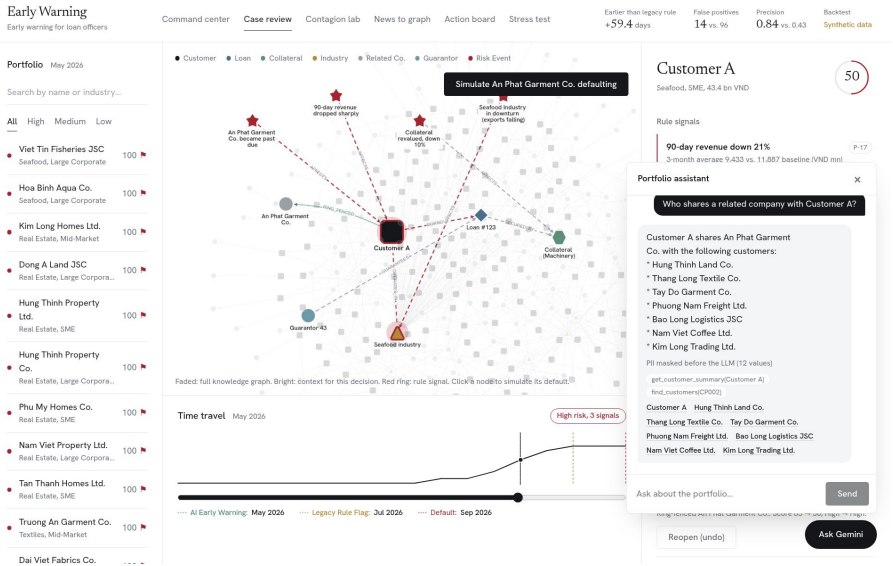
Contagion Matrix



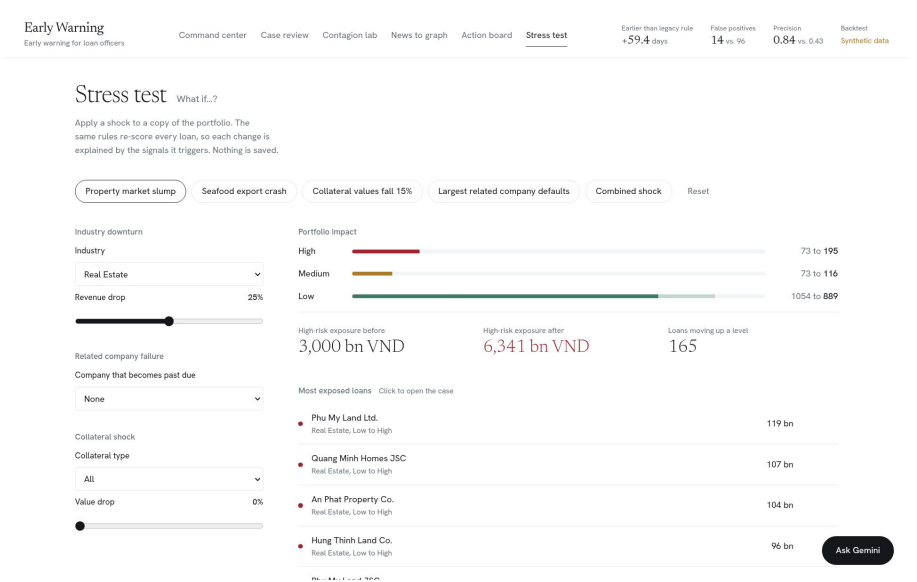
News to Graph



Action Board



Ask Gemini



Stress test

EARLY WARNING • BACKTEST & PERFORMANCE

Earlier signals. Faster decisions.

Detection quality

Metric	Early Warning	Legacy rule (DPD ≥ 30)
Defaulters caught	73/73 (100%)	73/73 (100%)
Warning before default	120 days	61 days
False positives	14	96
Precision	0.84	0.43

Backtest on 1,200 synthetic loans.

Response time • full 1,200-loan graph

Operation	Median	Max
Context graph + rule signals, one loan	2 ms	2 ms
Re-score whole portfolio	49 ms	49 ms
Stress test	55 ms	66 ms
Default cascade simulation	367 ms	445 ms
Complete a task → re-score everyone	86 ms	86 ms
Gemini news → graph event	1.4 s	1.6 s
Gemini portfolio assistant	3.1 s	3.1 s
Gemini case analysis (fresh / cached)	5.1 s / 4 ms	5.2 s / 7 ms

Median of 3–5 HTTP runs on a developer laptop; Gemini via Gemini API.

Every non-AI interaction responds in under 0.5 s, so the contagion simulation runs live on every click; Gemini answers in 1.4–5 s, and no score ever comes from the LLM.

Future Development

1



Richer graph

Add payment transactions, supply chains, shared directors and credit bureau (CIC) data as contagion channels.

2



Live news ingestion

Stream news and public registries through Pub/Sub, with Gemini extracting events continuously.

3



Production hardening on Google Cloud

Store the live state in Firestore, credentials in Secret Manager, and officer identities through Firebase Auth or IAP, with BigQuery for portfolio analytics.

1 — 2 — 3

4



Learning loop

Use officers' approve and reject decisions plus measured task outcomes to rank which mitigations work best.

5



Regulatory reporting

Generate Gemini-written early-warning reports for credit committees, with every figure traceable to the graph.

4 — 5

Provide links to your:

- GitHub Public Repository: https://github.com/Hu2Hoang/EWS
- Demo Video Link (3 Minutes): https://drive.google.com/file/d/1InDYW293xSuaFpzNTwimwzQQpNzx2vnQ/view?usp=drive_link
- Final Product Link: https://early-warning-420247575616.asia-southeast1.run.app

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Thank you

